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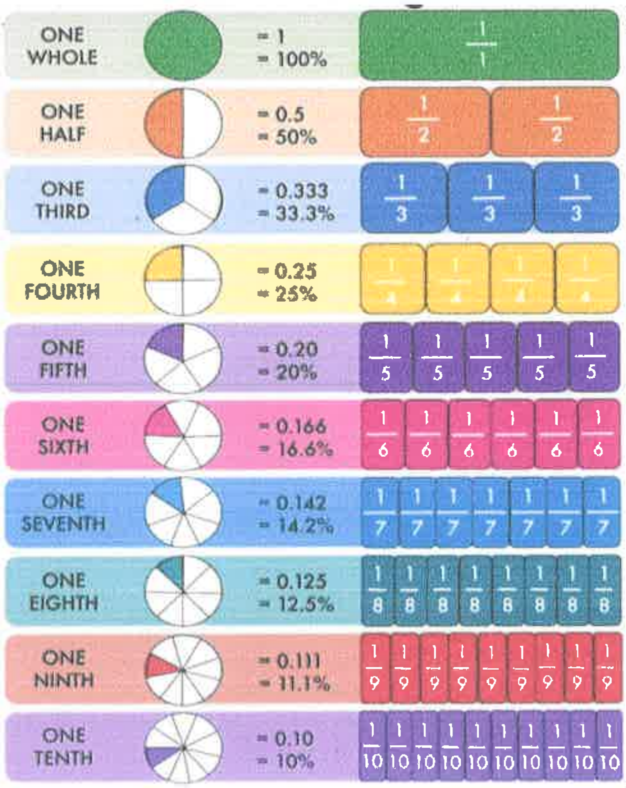

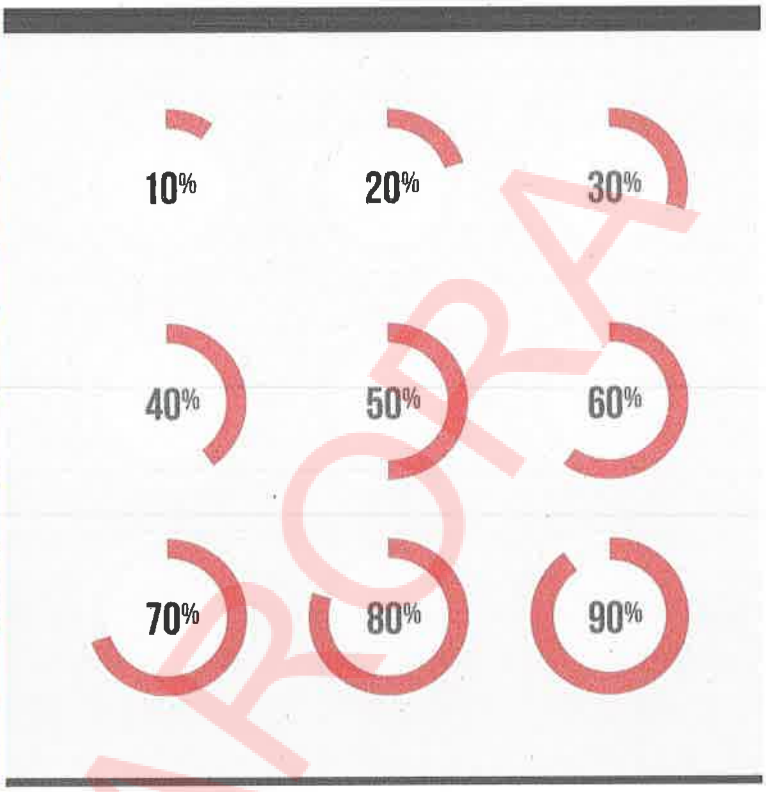

* $14.28\% = \frac{1}{7}$

Initial : Final

$7 \xrightarrow{+1\ (14.28\%\ \text{of}\ 7)} 8$

$\boxed{7 : 8}$

* $700 \xrightarrow[+14.28\%\ \text{of}\ 700]{+100} 800$

$\cancel{700} : \cancel{800}$
$7 : 8$

$\frac{700 \times 1}{7} = 100$

* $1400 \xrightarrow[+14.28\%\ \text{of}\ 1400]{+200} 1600$

$14\cancel{00} : 16\cancel{00}$
$14 : 16$
$7 : 8$

$\frac{1400 \times 1}{7} = 200$

* $770 \xrightarrow[+14.28\%\ \text{of}\ 770]{+110} 880$

$77\cancel{0} : 88\cancel{0}$
$77 : 88$
$7 : 8$

$\frac{770 \times 1}{7} = 110$

Decrease :-

$14.28\% = \frac{1}{7}$

Initial : Final

7 → 6

-1 (14.28% of 7)

$\boxed{7:6}$

* 840 $\xrightarrow[\text{– 14.28\% of 840}]{-120}$ 720 $\left(840 \times \frac{1}{7} = 120\right)$

$\left\{\begin{matrix} 84\cancel{0} : 72\cancel{0} \\ 84 : 72 \\ 7 : 6 \end{matrix}\right\}$

* 560 $\xrightarrow[\text{– 14.28\% of 560}]{-80}$ 480 $\left(560 \times \frac{1}{7} = 80\right)$

$\left\{\begin{matrix} 56\cancel{0} : 48\cancel{0} \\ 56 : 48 \\ 7 : 6 \end{matrix}\right\}$

* 490 $\xrightarrow[\text{– 14.28\% of 490}]{-70}$ 420 $\left(490 \times \frac{1}{7} = 70\right)$

$\left\{\begin{matrix} 49\cancel{0} : 42\cancel{0} \\ 49 : 42 \\ 7 : 6 \end{matrix}\right\}$

Find the ratio of Initial and Final Number →

+ ⇒ Increase

– ⇒ Decrease

$\boxed{I : F}$

- + 25% (1/4) → 4 : 5 (+1)
- – 25% (1/4) → 4 : 3 (–1)
- + 12.5% (1/8) → 8 : 9 (+1)
- – 12.5% (1/8) → 8 : 7 (–1)

$\boxed{I : F}$

- + 28.56% (2/7) → 7 : 9 (+2)
- – 28.56% (2/7) → 7 : 5 (–2)
- + 16.66% (1/6) → 6 : 7 (+1)
- – 16.66% (1/6) → 6 : 5 (–1)

- $+25\% \Rightarrow \begin{matrix} I : F \\ 4 : 5 \end{matrix} \Rightarrow$ Also, $\begin{matrix} I : F \\ 1 : \frac{5}{4} \end{matrix}$

$\Rightarrow \frac{I}{F} = \frac{4}{5}$ $\boxed{\frac{5}{4} I = F}$

These are the ways of Representation

- $-25\% \Rightarrow \begin{matrix} I : F \\ 4 : 3 \end{matrix} \Rightarrow$ Also, $\begin{matrix} I : F \\ 1 : \frac{3}{4} \end{matrix}$

$\Rightarrow \frac{I}{F} = \frac{4}{3}$ $\boxed{\frac{3}{4} I = F}$

- $+20\% \Rightarrow \frac{1}{5}$ $\begin{matrix} I : F \\ 5 : 6 \end{matrix} \Rightarrow F = I \times \frac{6}{5}$ $\left(1 + \frac{1}{5} = \frac{6}{5}\right)$
- $+10\% \Rightarrow \frac{1}{10}$ $\begin{matrix} I : F \\ 10 : 11 \end{matrix} \Rightarrow F = I \times \frac{11}{10}$ $\left(1 + \frac{1}{10} = \frac{11}{10}\right)$
- $+14.28\% \Rightarrow \frac{1}{7}$ $\begin{matrix} I : F \\ 7 : 8 \end{matrix} \Rightarrow F = I \times \frac{8}{7}$ $\left(1 + \frac{1}{7} = \frac{8}{7}\right)$
- $+16.66\% \Rightarrow \frac{1}{6}$ $\begin{matrix} I : F \\ 6 : 7 \end{matrix} \Rightarrow F = I \times \frac{7}{6}$ $\left(1 + \frac{1}{6} = \frac{7}{6}\right)$

- $+25\% \Rightarrow 1 + \frac{1}{4} = \frac{5}{4} \Rightarrow I + \frac{I}{4} = F$

$I\left(1 + \frac{1}{4}\right) = F$

When Initial Number is increased by 25%

* $\boxed{I \times \frac{5}{4} = F}$

- $+6.25\% \Rightarrow \frac{1}{16}$ | I : F, 16 : 17 | $I \times \frac{17}{16} = F$
- $+6.66\% \Rightarrow \frac{1}{15}$ | I : F, 15 : 16 | $I \times \frac{16}{15} = F$
- $+11.11\% \Rightarrow \frac{1}{9}$ | I : F, 9 : 10 | $I \times \frac{10}{9} = F$
- $+9.09\% \Rightarrow \frac{1}{11}$ | I : F, 11 : 12 | $I \times \frac{12}{11} = F$
- $+25\% \Rightarrow \frac{1}{4}$ | I : F, 4 : 5 | $I \times \frac{5}{4} = F$
- $-6.25\% \Rightarrow \frac{1}{16}$ | I : F, 16 : 15 | $I \times \frac{15}{16} = F$
- $-6.66\% \Rightarrow \frac{1}{15}$ | I : F, 15 : 14 | $I \times \frac{14}{15} = F$
- $-11.11\% \Rightarrow \frac{1}{9}$ | I : F, 9 : 8 | $I \times \frac{8}{9} = F$
- $-9.09\% \Rightarrow \frac{1}{11}$ | I : F, 11 : 10 | $I \times \frac{10}{11} = F$
- $-25\% \Rightarrow \frac{1}{4}$ | I : F, 4 : 3 | $I \times \frac{3}{4} = F$
- $+10\% \Rightarrow \frac{1}{10}$ | I : F, 10 : 11 | $I \times \frac{11}{10} = F$
- $-10\% \Rightarrow \frac{1}{10}$ | I : F, 10 : 9 | $I \times \frac{9}{10} = F$

- Increase 480 by 25% $\Rightarrow$ $25\% = \frac{1}{4}$

1) $480 + 120 = 600$

2) $480 \times \frac{5}{4} = 600$

$\left\{1 + \frac{1}{4} = \frac{4+1}{4} = \frac{5}{4}\right\}$

- Increase 600 by 20% $\Rightarrow$ $20\% = \frac{1}{5}$

$600 \times \frac{6}{5} = 720$

- Increase 500 by 10% $\Rightarrow$ $10\% = \frac{1}{10}$

$50\cancel{0} \times \frac{11}{1\cancel{0}} = 550$

- Increase 880 by 12.5% $\Rightarrow$ $12.5\% = \frac{1}{8}$

$880 \times \frac{9}{8} = 990$

- Increase 666 by 16.66% $\Rightarrow$ $16.66\% = \frac{1}{6}$

$666 \times \frac{7}{6} = 777$

* Ways of Representation $\rightarrow$

$\Rightarrow$ $\frac{a}{b} : \frac{c}{d}$

$ad : cb$

$\Rightarrow$ $\frac{3}{5} : \frac{4}{7}$

$(3\times7) : (5\times4)$

$21 : 20$

$\Rightarrow$ $5 : 7$

$(5\times1) : 7$

$1 : \frac{7}{5}$

$\Rightarrow$ $\frac{9}{11} : \frac{3}{5}$

$(9\times5) : (11\times3)$

$45 : 33$

$15 : 11$

A bag weighs 80 kg. It is increased by 25%. Find the new weight.
एक बैग का वजन 80 kg है। इसमें 25% की वृद्धि की जाती है। नया वजन ज्ञात कीजिए।

1) $80 \xrightarrow[\text{+ 25\% of 80}]{+20} 100$

2) $F = I \times \frac{5}{4}$ $F = 80 \times \frac{5}{4}$ $\left(1 + \frac{1}{4}\right)$

$F = 100$

3) $I : F$

$4 : 5$

$\times 20$ $\times 20$

$80 : 100$

These are the methods or alternatives to solve the question.

The price of a pen is 40. It is decreased by 10%. Find the new price.
एक पेन की कीमत 40 है। इसकी कीमत में 10% की कमी की जाती है। नई कीमत ज्ञात कीजिए।

1) $40 \xrightarrow[\text{− 10\% of 40}]{-4} 36$

2) $F = I \times \frac{9}{10}$

Decrease by 10%

$F = 40 \times \frac{9}{10}$

$= 36$

3) $I : F$

$\cancel{10} : 9$ (Decrease by 10%)

$\times 4$

$4\cancel{0}$ 36

A student scored 160 marks. Marks are decreased by 12.5%. Find the new marks.
एक छात्र ने 160 अंक प्राप्त किए। अंकों में 12.5% की कमी की जाती है। नए अंक ज्ञात कीजिए।

1) $160 \xrightarrow[\text{− 12.5\% of 160}]{-20} 140$ $\left(160 \times \frac{1}{8} = 20\right)$

2) $F = 160 \times \frac{7}{8}$

$F = 140$

3) $I : F$

$\cancel{8} : 7$

$\times 20$

$\cancel{160}$ 140

20

The weight of a packet increases from 48 kg to 60 kg. Find the percent increase.
एक पैकेट का वजन 48 kg से बढ़कर 60 kg हो जाता है। प्रतिशत वृद्धि ज्ञात कीजिए।

1) $48 \xrightarrow{+12} 60$

(Increased by 12)

$\frac{\cancel{12}}{\cancel{48}_4} \times 100 = \frac{\cancel{100}}{\cancel{4}}$

$= \underline{\underline{25\%}}$

3) I : F

$\cancel{48} : \cancel{60}$

4 : 5

+1

$\frac{1}{4} \times 100 = 25\%$

2) $48 \longrightarrow 60$

$\frac{\cancel{60}}{\cancel{48}} \; \frac{5}{4}$

$\frac{5}{4} - 1 = \frac{1}{4}$ or $\underline{\underline{25\%}}$

The marks of a student decrease from 96 to 84. Find the percent decrease.
एक छात्र के अंक 96 से घटकर 84 हो जाते हैं। प्रतिशत कमी ज्ञात कीजिए।

1) $96 \xrightarrow{-12} 84$

(Decreased by 12)

$\frac{\cancel{12}}{\cancel{96}_8} \times 100 = \frac{100}{8}$

$= \underline{\underline{12.5\%}}$

3) I : F

96 : 84

8 : 7

−1

$\frac{1}{8} \times 100 = \underline{\underline{12.5\%}}$

2) $96 \longrightarrow 84$

$\frac{\cancel{84}}{\cancel{96}} \; \frac{7}{8}$

$1 - \frac{7}{8} = \frac{1}{8}$

The number of employees decreases from 360 to 300. Find the percent decrease.
कर्मचारियों की संख्या 360 से घटकर 300 हो जाती है। प्रतिशत कमी ज्ञात कीजिए।

1) $360 \xrightarrow{-60} 300$

(Decreased by 60)

$\frac{6\cancel{0}}{36\cancel{0}} \times 100 = 16.66\%$

2) $360 \longrightarrow 300$

$\frac{30\cancel{0}}{36\cancel{0}} = \frac{5}{6}$

$1 - \frac{5}{6} = \frac{1}{6}$ or 16.66%

3) I : F

$36\cancel{0} : 30\cancel{0}$

$36 : 30$

$6 : 5$

-1

$\frac{1}{6} \times 100 = 16.66\%$

After a 16.67% increase, the marks become 105. Find the original marks.
16.67% की वृद्धि के बाद अंक 105 हो जाते हैं। मूल अंक ज्ञात कीजिए।

1) $16.67\% = \frac{1}{6}$

$+16.67\% = 1 + \frac{1}{6} = \frac{7}{6}$

$F = I \times \frac{7}{6}$

$105 = I \times \frac{7}{6} \Rightarrow \boxed{I = 90}$

2) I : F

$6 : \cancel{7}$

$\times 15$

90 $\quad \cancel{105}$ 15

A earns ₹ 56000 and B earns ₹ 40000. A's salary is how much percent more than B's salary?
A की आय ₹ 56000 है और B की आय ₹ 40000 है। A की आय, B की आय से कितने प्रतिशत अधिक है?

1) A : B

$56000 : 40000$

$+16000$

$\frac{16\cancel{000}}{40\cancel{000}} \times 10\cancel{0} = 40\%$

2) A : B

$56\cancel{000} : 40\cancel{000}$

$56 : 40$

$7 : 5$

$+2$

$\frac{2}{5} \times 100 = 40\%$

Q. B's salary is how much percent less than A's ?

A : B

$56\cancel{000} : 40\cancel{000}$

$\underbrace{7 : 5}_{-2}$

$\Rightarrow \frac{2}{7} \times 100 = 28.56\%$

Q. B's salary is what percent of A's salary ?

$\frac{5}{7} \times 100 \Rightarrow 71.42\%$

Q. A's salary is what percent of B's salary ?

$\frac{7}{5} \times 100 \Rightarrow 140\%$

Rahul scored 72 and Amit scored 90. Rahul's marks are how much percent less than Amit's marks?
राहुल ने 72 अंक और अमित ने 90 अंक प्राप्त किए। राहुल के अंक, अमित के अंकों से कितने प्रतिशत कम हैं?

Rahul : Amit

$\cancel{72} : \cancel{90}$

$\underbrace{4 : 5}_{-1}$

$\frac{1}{5} \times 100 \Rightarrow 20\%$

Base value will be Amit's marks.

Two quantities are 64 and 24. Their difference is what percent of the smaller quantity?
दो राशियाँ 64 और 24 हैं। उनका अंतर, छोटी राशि का कितने प्रतिशत है?

Quantity 1 : Quantity 2

$\cancel{64} : \cancel{24}$

$8 : 3$

5 ← Difference

Difference

$\frac{5}{3} \times 100$

smaller Qty.

$\left(\frac{3}{3} + \frac{2}{3}\right) \Rightarrow 1 + \frac{2}{3}$

100% 66.66%

$\frac{5}{3} \Rightarrow 166.66\%$

A is 24 and B is 64. By what percent should A increase to become equal to B?
A का मान 24 है और B का मान 64 है। A को B के बराबर होने के लिए कितने प्रतिशत बढ़ाना होगा?

A : B
$\cancel{24} : \cancel{64}$
3 : 8
+5

{ 3 should be increased by 5 to become equal to 8 }

$\frac{5}{3} \times 100 \Rightarrow \left(\frac{3+2}{3}\right) \Rightarrow 1 + \frac{2}{3}$

$\Rightarrow 166.67\%$

Two students scored 84 and 105. By what percent should 84 increase to become equal to 105?
दो छात्रों ने 84 और 105 अंक प्राप्त किए। 84 को 105 के बराबर होने के लिए कितने प्रतिशत बढ़ाना होगा?

Student 1 : Student 2
$\cancel{84} : \cancel{105}$
28 : 35
+7

{ 28 should be increased by 7 to become 35 }

$\frac{\cancel{7}^{1}}{\cancel{28}_{4}} \times 100 = \frac{1}{4} \times 100$

$= 25\%$

A number is increased by 25% to become 18,375. Find the original number.
एक संख्या को 25% बढ़ाने पर उसका मान 18,375 हो जाता है। मूल संख्या ज्ञात कीजिए।

(A) 13,500 (B) 14,500 (C) 15,000 (D) 14,700 (E) 12,700

1) $25\% = \frac{1}{4}$

$+25\% = 1 + \frac{1}{4} = \frac{5}{4}$

$F = I \times \frac{5}{4}$

$18375 = I \times \frac{5}{4} \Rightarrow \boxed{I = 14700}$

2) I : F
4 : $\cancel{5}$
×3675
14700 $\cancel{18375}$ 3675

A number is increased by 15.38% and the resultant number is 3,225. What is the original number?
एक संख्या को 15.38% बढ़ाने पर प्राप्त संख्या 3,225 है। मूल संख्या ज्ञात कीजिए।

(A) 2,975 (B) 2,925 (C) 2,795 (D) 2,800 (E) 2,745

1) $15.38\% = \frac{2}{13}$

$+15.38\% = 1 + \frac{2}{13} = \frac{15}{13}$

$3225 = I \times \frac{15}{13}$

$I = \frac{3225 \times 13}{15}$

$\boxed{I = 2795}$

2) I : F

13 : 15

×215

2795   3225

If 4,158 is decreased by 16.67%, what is the resulting number?
यदि 4,158 में 16.67% की कमी की जाए, तो प्राप्त संख्या क्या होगी?

(A) 3,450 (B) 3,465 (C) 3,465.5 (D) 3,665 (E) 3,900

1) $4158 \xrightarrow[-16.67\% \text{ of } 4158]{-693} 3465$

$\left(4158 \times \frac{1}{6} = 693\right)$

2) $F = I \times \frac{5}{6}$

$F = 4158 \times \frac{5}{6}$

$F = 3465$

3) I : F

$\cancel{6}$: 5

×693

$\cancel{4158}$ 3465

693